

ers around it—to prevent the indium atoms from dissociating from the other atoms. Such dissociation can form areas of inhomogeneous indium alloys, or islands, which, in turn, causes the electron-hole recombination energy to vary from point to point. That variation makes the emission spectrum too broad to yield the coherent, monochromatic light expected from a laser. Thus when the technicians raise the reactor temperature to grow the next GaN layer atop the delicate InGaN layer just deposited, they must be very careful so as not to form too many of these islands.

In the case of polar diodes using 'c'-plane sapphire substrate, the strong internal fields force the manufacturers to make the active layers of InGaN exceedingly thin—less than 4 nm or only about 20 atoms thick. This approach helps to keep electrons and holes huddled together closely, boosting the chances that they will meet to create light.

Because non-polar and semipolar diodes have internal electric fields that are almost negligible, their InGaN active layers can be grown substantially thicker up to 20 nm. The indium islands still form in these robust layers, but closer to the interfaces with the surrounding GaN. Confining the islands to the interfaces boosts the chances of getting the narrower light spectrum needed for laser action. Moreover, the thicker robust layers help to simplify manufacturing process as extra 'cladding' layers in the diode stack to help trap and guide the photons can be eliminated. In the case of 'c'-plane geometry technique, provision of extra 'cladding' layers is a necessity.

An elegant approach to single-frequency laser operation that has found widespread application involves the use of distributed resonators, fabricated into the laser structure to give integrated wavelength selectivity. The operation of this type of laser is based on the distributed Bragg phase-grating reflector (DBR). As shown in Fig.

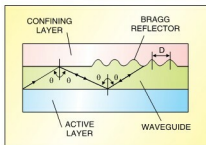


Fig. 5: Periodic waveguide structure

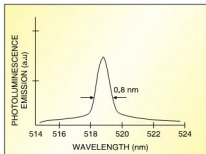


Fig. 6: Emission spectra of green InGaN MQW laser

3(a), a DBR consisting of 13.5 pairs of $\text{Ta}_2\text{O}_5/\text{SiO}_2$ layers is deposited onto the surface of both p-GaN and n-GaN. The reflectivity of the DBR is about 99.2 per cent at centre wavelength.

A phase grating is essentially a region of periodically varying refractive index that causes two counter-propagating travelling waves to couple (Fig. 5). When light propagating in the guide impinges on the corrugation, some of the energy is diffracted out of the waveguide into either the cover or substrate. Light satisfying the Bragg condition is reflected back along the guide at 180° to the original direction of propagation. By providing a periodic perturbation in thickness or refractive index along the direction of propagation, one can achieve a highly selective wavelength reflection. Such a wavelength-selective reflection can be used to provide low-loss feedback at only one oscillation frequency and higher losses at other frequencies.

When period of corrugation 'D' is equal to $K\lambda_b/2n_g$, where 'K' is the integer order of grating, ' λ_b ' is the Bragg wavelength and ' n_g ' is the effective refractive index of the waveguide, the

mode near Bragg wavelength ' λ_b ' is reflected constructively (called Bragg reflection). Therefore this particular mode lasts whilst the other modes exhibiting higher losses are suppressed from oscillation.

High-power DBR lasers are used in applications requiring single-frequency and wavelength tunability over a few tenths of a nanometre for generation of visible light by harmonic frequency conversion. The ability to fabricate these devices in a single epitaxial growth step with grating defined by holographic interferometry significantly reduces the device complexity and processing time, thus minimising manufacturing cost.

Using semipolar geometry, green LEDs that operate at 519nm wavelength and achieve quantum efficiencies (the ratio of photons emitted to electrons flowing in) above 45 per cent have been manufactured recently. Emission spectrum at 40°C of a typical green laser oscillating at 519 nm is shown in Fig. 6.

The main problem now is the availability of a large enough substrate at acceptable cost. To produce green laser diodes, manufacturers should have GaN non-polar and semipolar substrate at least 5 cm in diameter costing about Rs 80,000 per wafer.

Applications

Future models that rely on green lasers for handheld projectors allow for much brighter and efficient displays, and also shrink the projectors enough to allow them to fit inside the cell phone. Inside the laser projector, red, blue and green lasers focus onto a single mirror about the size of a pinhead. As light bounces off the reflector, the mirror assembly rapidly scans back and forth to project pixels one by one onto a screen or wall. No use of lens means the projector never needs to be focused. The projector has DVD-equivalent resolution of 848×480 pixels. ●

The author is a regular contributor to EPC